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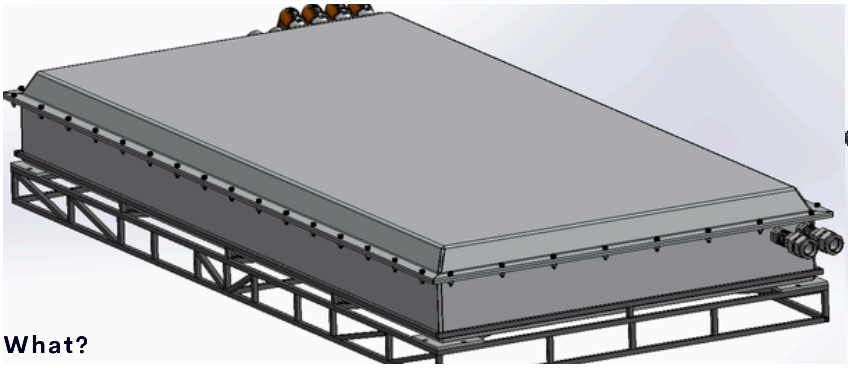
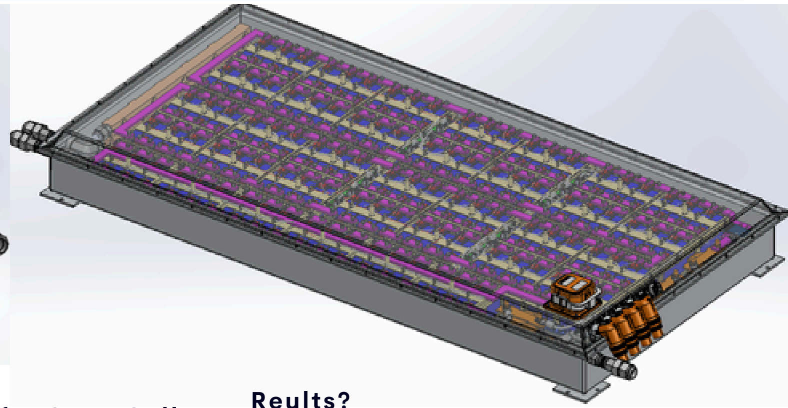
HIGH VOLTAGE ELECTRIC VEHICLE BATTERY PACK - OHMITRON, INC





What?

- Focused on development and production of **900V High Voltage Battery Pack (HVBP), promoting Electricification** of commercial, transit and heavy-duty vehicles.
- Integration of **Innovative Cooling** Technology for BPs, reducing Fire Hazards and Failures in EVs
- Used Composite Materials, Polymer and Metal Structures

How?

- Utilized **SolidWorks 2023** for **CAD**, **Cell arrangement**, **thermal management system**, **sheet metal**, **weldments** and **frame design**
- Utilized **GD&T** for **DFA/DFM**, **SolidWorks Electrical** for **electrical harness/cable design,** integration of **Battery Management Systems (BMS)** and **Electrical Design**.
- Conducted rigorous **ANSYS simulations**, **FEA** and **CFD** for **thermal analysis/runaway**.
- Co-op with Electrical Team for Electrical schematics, components and connection.

Reults?

- Significantly **improved operational cycles** with **reduced charging time** to **20 minutes** at a **10C-rate**.
- Successfully integrated innovative cooling technology and a **35kWhr**, **840V HVBP** with controlled Thermals
- Achieved **IP65** rating and **HVIL** integration.
- Proposed and Patented in **DOT SBIR** and American Patents Respectively

BATTERY PACK ELECTRONICS (FIG.1), BATTERY PACK CELL MODULE (FIG.2), BATTERY PACK CELL HOLDER FEA ANALYSIS (FIG.3), BATTERY PACK THERMAL/FLOW ANALYSIS (FIG.4)

Fig. 1

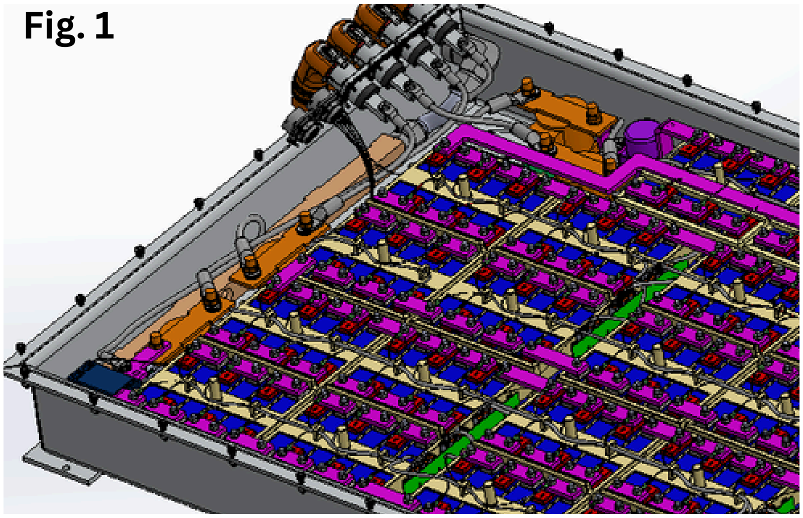
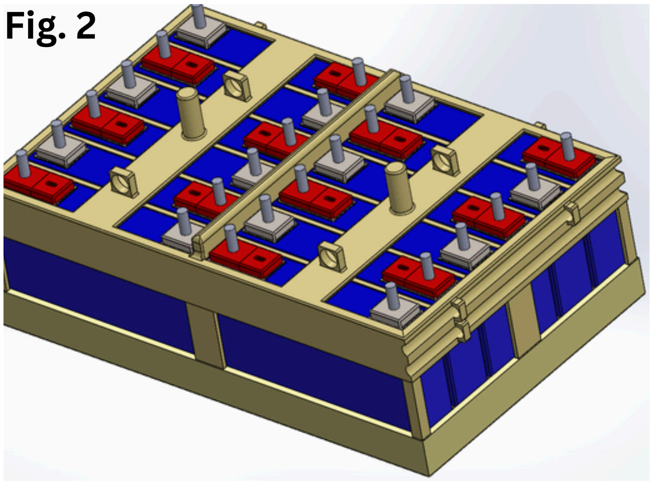


Fig. 2



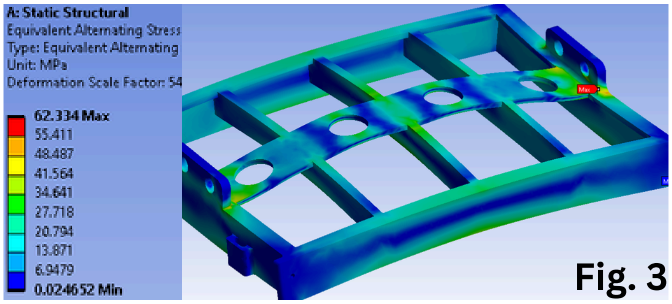


Fig. 3

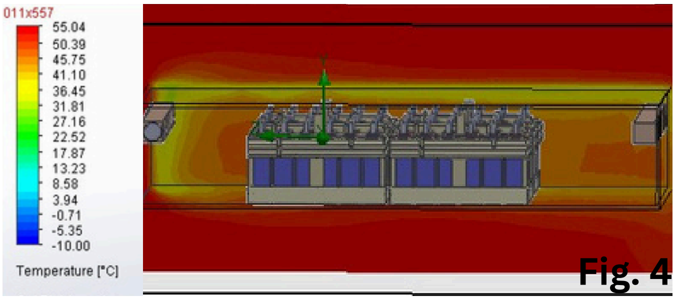
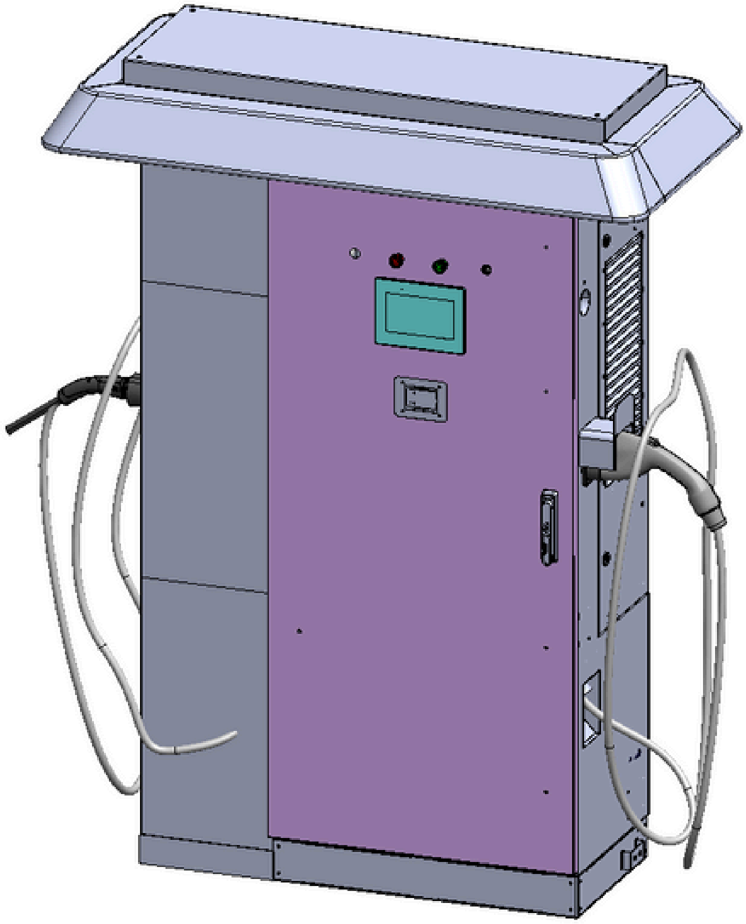
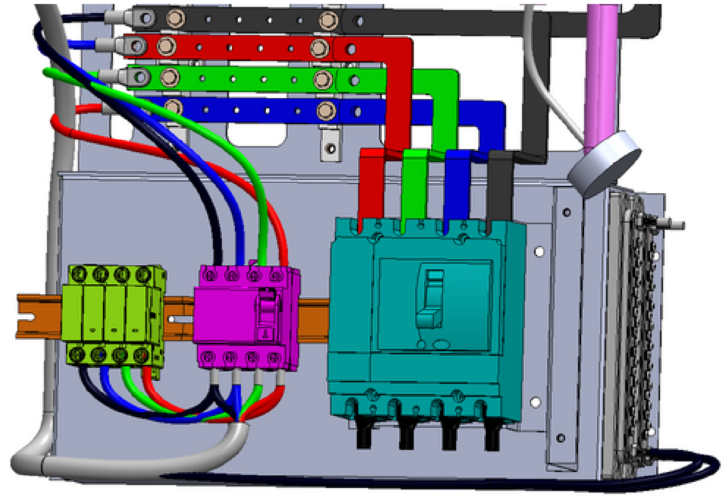


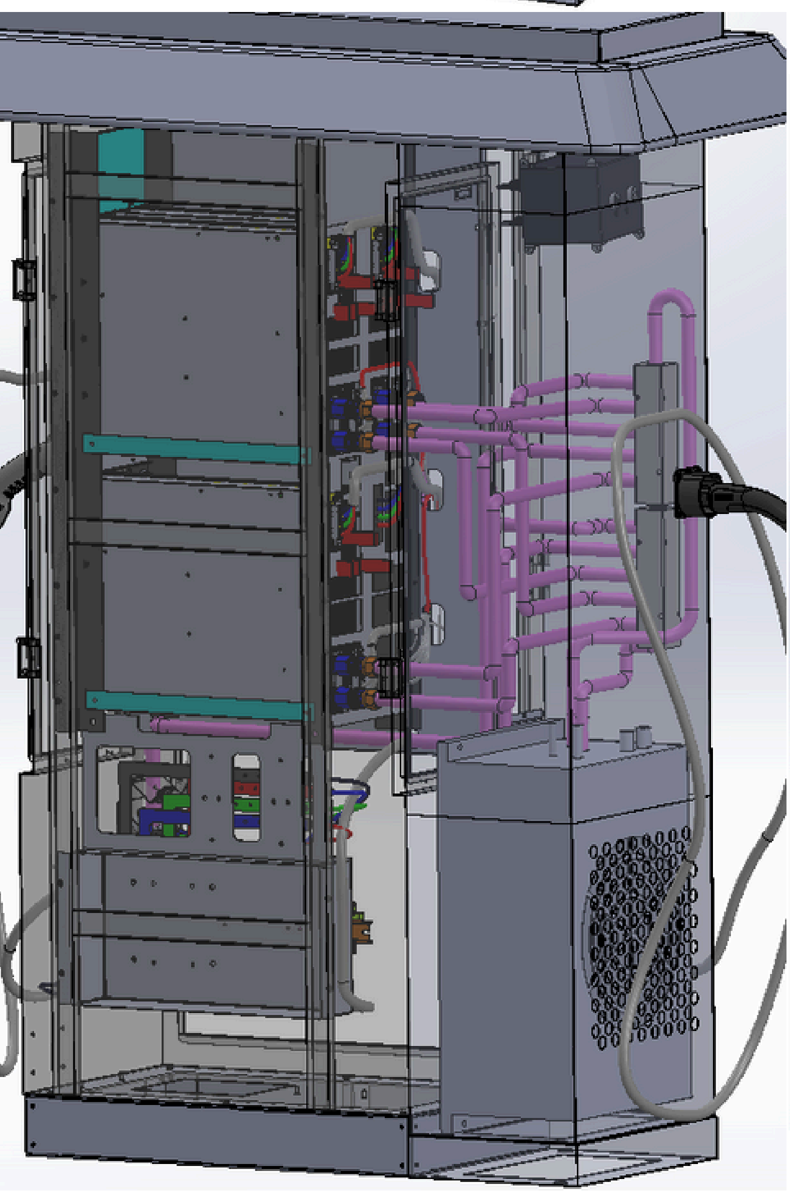
Fig. 4

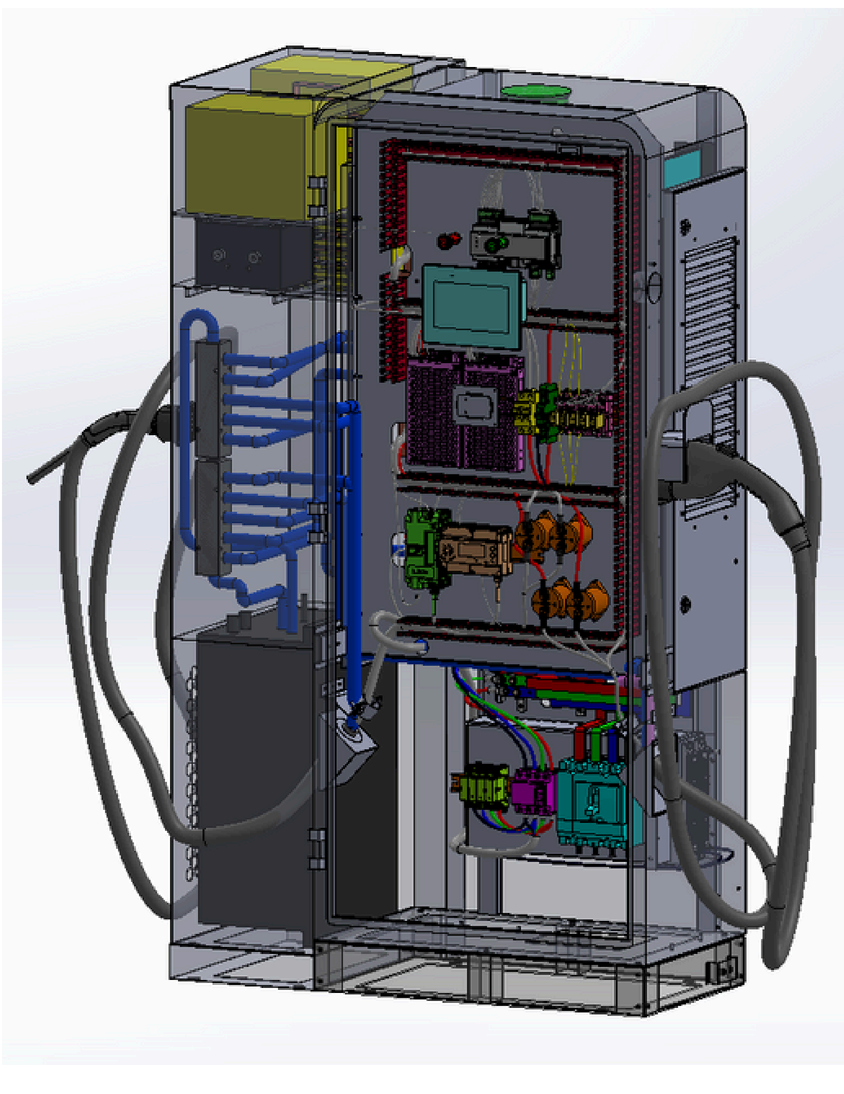




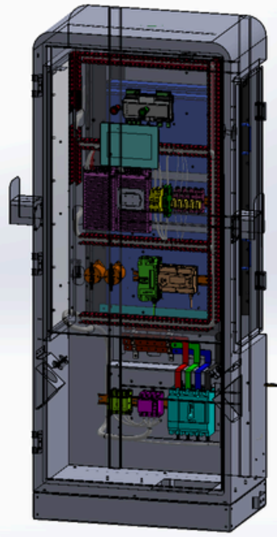
Ohmitron’s Propietrary Electric Vehicle Charger

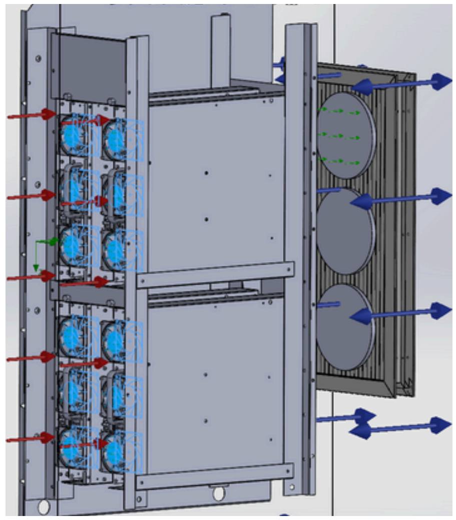
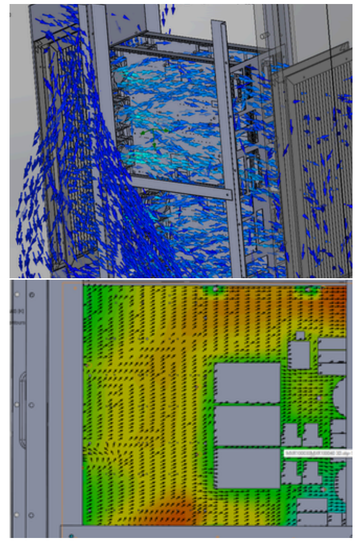
(Detailed assembled industrial design)





ELECTRIC VEHICLE 360KW CHARGER - OHMITRON, INC



What?

- Developed a **versatile convertible EV Charger** delivering **120kW**, 180**kW**, 200**kW**, and **360W** power outputs.
- The project yeilded a standardized **120kW** scalable EV Charger just by **adding or removing AC/DC converter modules**.
- Compatible with **CCS1** and **NACS charging systems** incorporating **Intelligent charging controller** within a robust **sheet metal cabinet**.

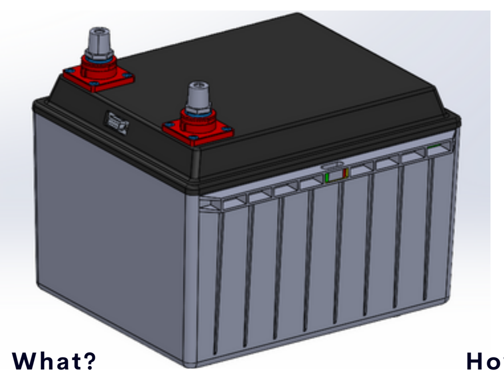
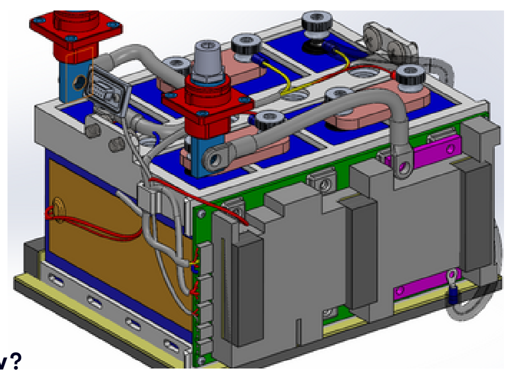
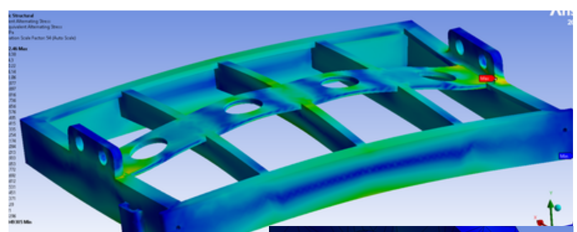
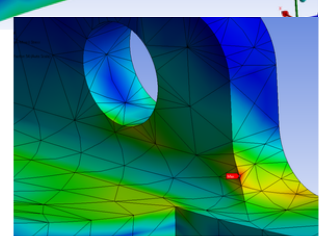
How?

- Developed EV Charger Cabinet using **Solidworks** for **CAD**, **1.5mm** thick **sheet metal designing** applying **GD&T**
- Cabinet structure optimization according to **DFA/DFM** by **structural FEA**, **thermal** and **airflow CFD analyses** using **Ansys**.
- Used **Solidworks Electrical** for **Electrical Design** and **Cabling**
- Applied **HVAC** and **Thermo-mechanics**
- Co-op with Electrical and Software teams for integration of components

Reults?

- Successfully achieved the objectives of **scalability** of standardized **120kW** design with its modular architecture.
- Provided an efficient, **cost-effective** solution for variety of Electric vehicle's
- Achieved optimal temperature conditions (**55-65C** on max load)
- Efficiently managed Thermal Losses and Air Cooling.

AUTOMOBILE GRADE LITHIUM ION 12V BATTERY - OHMITRON, INC

What?

- Development of **Automobile Grade 12V Lithium-ion battery**
- Incorporated **4 prismatic li-ion** cells, each rated at 3.2V adaptable design across various ampere ratings (50A, 70A, 100A, 200A, 240A, 280A, 400A and 480A)
- Achieved **automotive-grade IP65** rating for vehicles, though versatile for use in household **power storage, solar systems** and various vehicles such as trucks, boats and others.

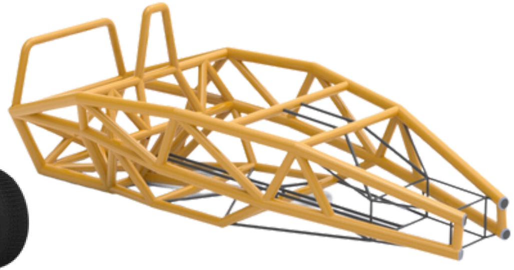
How?

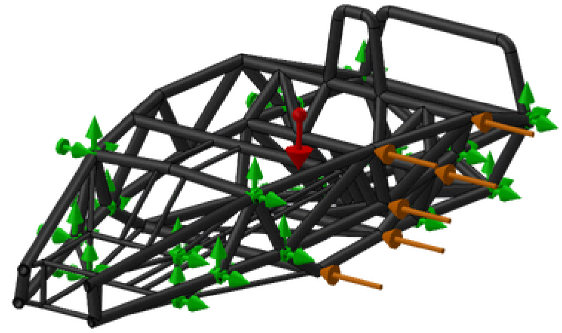
- **Design Standardization** done by making battery's core design **modular**, enabled compatibility across various applications
- **Casing** and **Cell Holders** engineered to withstand high loads, vibrations, impacts and wear, featuring **mounts** for **HV Headers**, **CANBus** and **BMS** according to DFA/DFM using **GD&T**
- By **Innovative cooling** and **heating pads Thermal Management** ensured the **battery** and **BMS** operational **extreme cold** Temps of **-40°C, far below** the typical **-20°C limit for Li-ion systems**
- Used **Solidworks** for **CAD** optimization and **ANSYS** for **FEA, Modal/Frequency, thermal CFD and impact analysis** of cell holders and casing

Results?

- Successfully prototyped the 12V Lithium ion batteries
- Used the **Innovative Cooling** technology and **operated BMS** and **Battery** at **-40C** using advanced cells and **Thermal Pads** beyond typical **-20C** limit
- Achieved almost **Infinite life** for Cell Holders by **FEA** and **Fatigue Design Life Analysis**
- **Electrical Design** and **Cable management**

TWICE 1ST POSITION IN XRC AUTONOMOUS VEHICLE - ASME E-FESTS 2023





What?

- ASME XRC AV **Global Competition** at **ASME E-Fests** in **2023** and **2024**
- **Designing** of **Ariel Atom Vehicle** for **2023** and **Mercedez Formula 1** for **2024** within Size, Dimensions and Weight **Constraints**
- Using Block Coding to make it fully **Autonomous** and Race it on any **Racetrack**

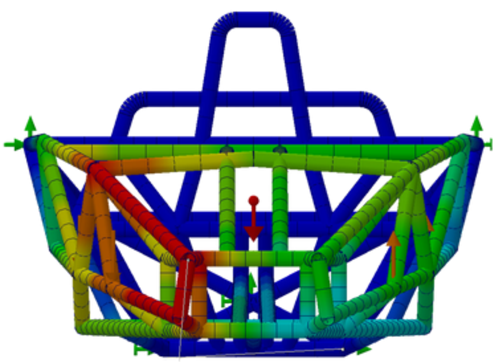
How?

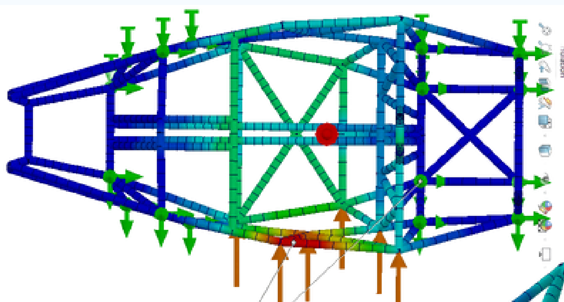
- Used **Solidworks** for Designing of whole **Chasis** using **Blueprints**
- Used **Weldment**s For Chasis and **Surface Modelling** for **Spoiler** and other Features

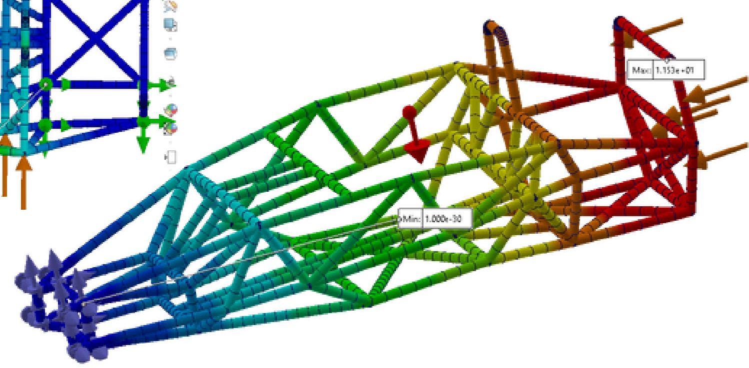
Reults?

- Secured **1st position, twice,** in the Competition **Worldwide** by Fastest Finishing time of **15** and **13** sec in **2023** and **2024** respectively
- Keeping **low weight profile** of vehicle and Reduced Drag **increasing Speed** by **15%** then others

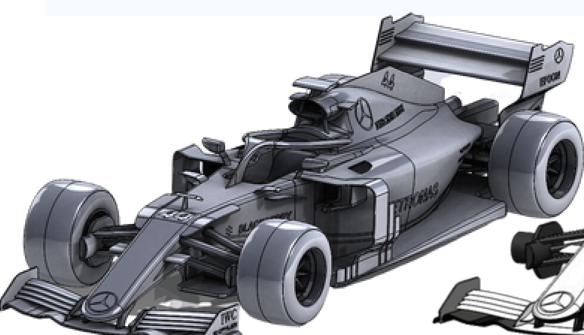
FINITE ELEMENT ANALYSIS FOR CRASHING SCENARIOS WITH RACETRACK WALLS OR OTHER VEHICLES

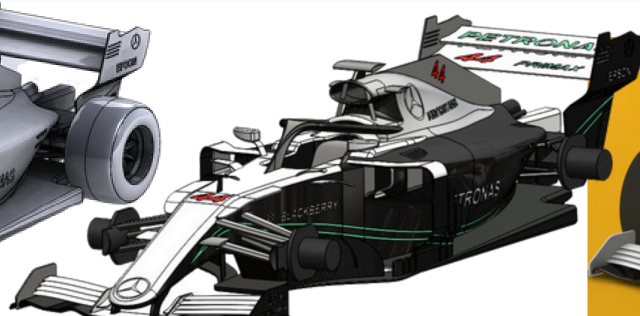






MERCEDEZ F1 VEHICLE - DASSAULT SYSTEMS X ASME E-FESTS 2024





What?

- Recreated a **precise** and **intricate** model of the **Mercedes Formula 1** vehicle for **Dassault Systems**
- Showcasing **advanced 3D modeling skills** using **Solidworks 2023**
- Used the Vehicle Model in **ASME E-Fests XRC AV Challenge** in **2024**

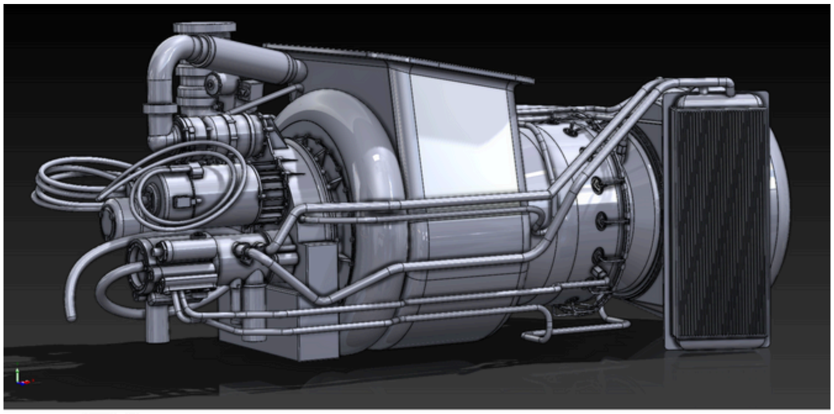
How?

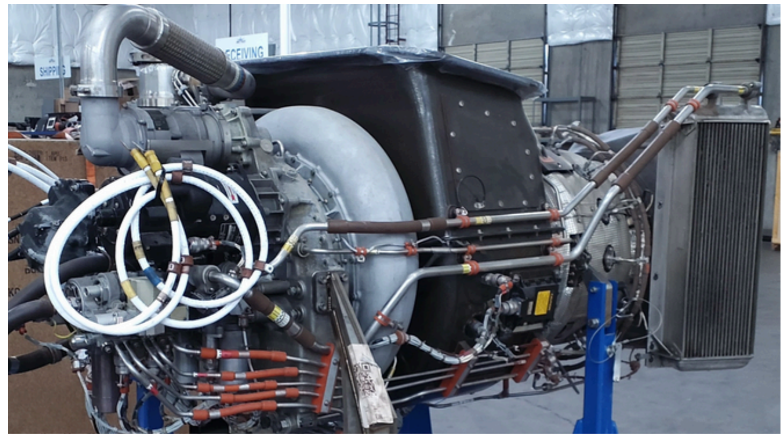
- Employed **surface modeling** and advanced methods to create every detail of the racing machine in **Solidworks 2023**
- Applied **CAD Design Optimization** to reduce weight and Drag for Competition as well

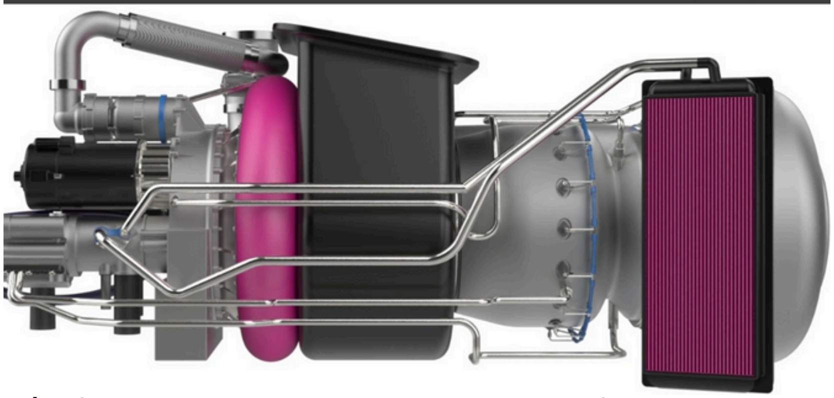
Results?

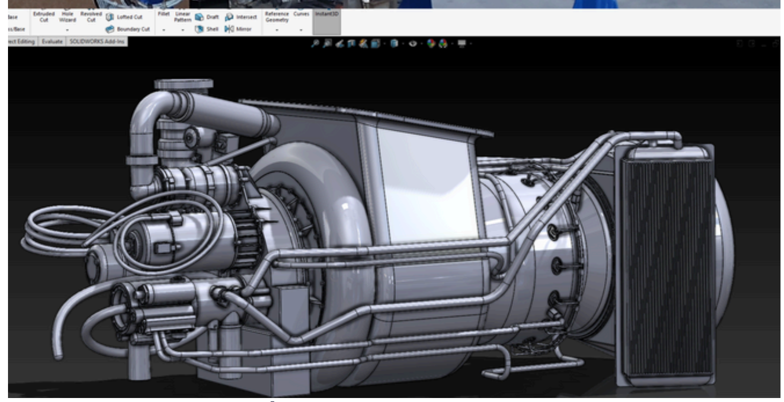
- Secured **1st position** in the Competition **Worldwide** by Fastest Finishing time of **13** sec in **2024**
- Dassault Systems **Showcased** The Vehicle on **Media Platforms appreciating Mentioning my Work**

CAD DESIGN OF GTCP 131-9C & GTCP 36-150RJ APUS - DUNIT SOL









What?

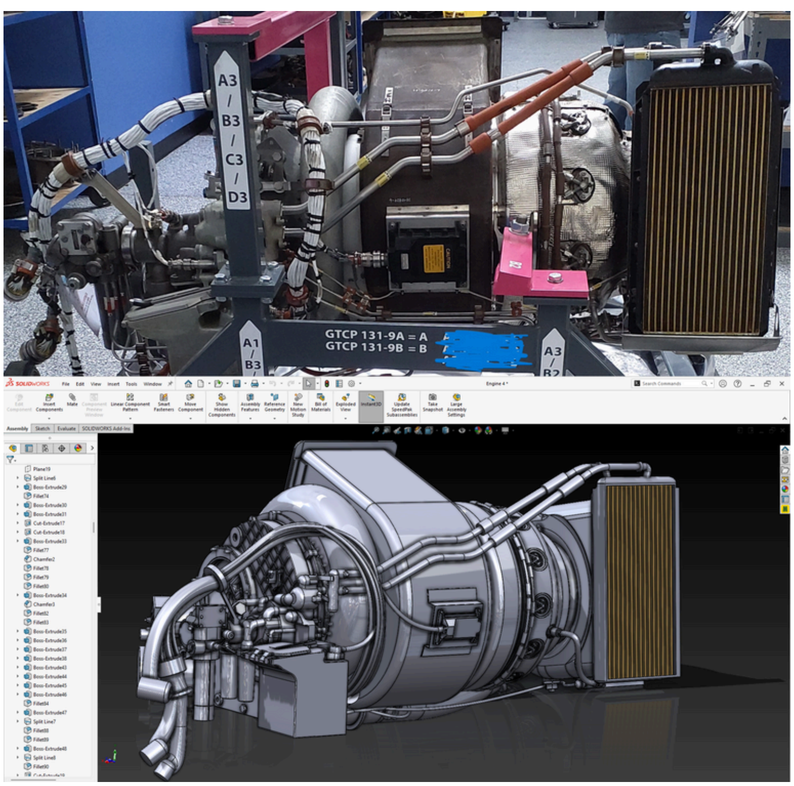
- **Undertook** the **comprehensive design** of **GTCP 36-150RJ** and **131-9 APUs (Auxiliary Power Unit)** using **Solidworks 2023**.
- These projects aimed to create a **precise digital design** of this **critical aircraft components,** used by **Honeywell**.

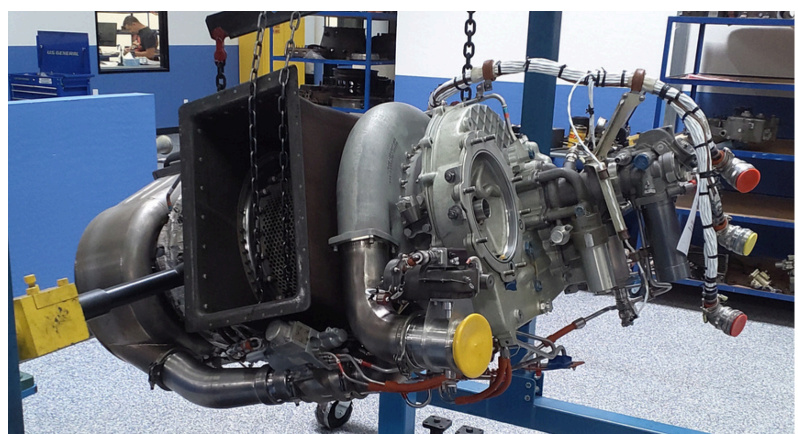
How?

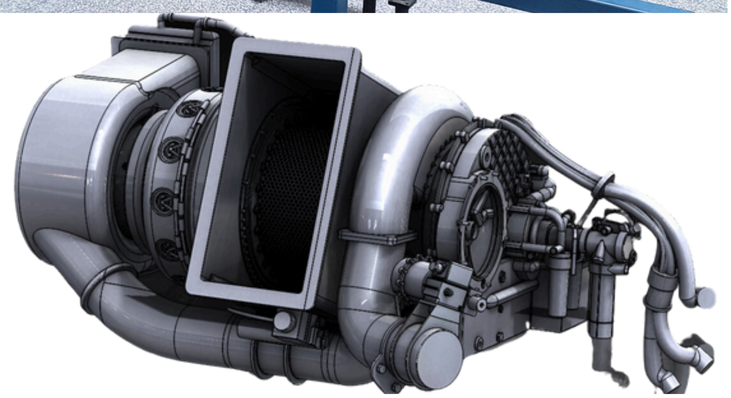
- **Collaborating closely** with the client, I meticulously integrated all functional aspects of the APUs while **adhering to specific dimension constraints** and **aspect ratios** provided.
- The process involved **translating reference images of the APUs** from various angles into **detailed CADs.**

Results?

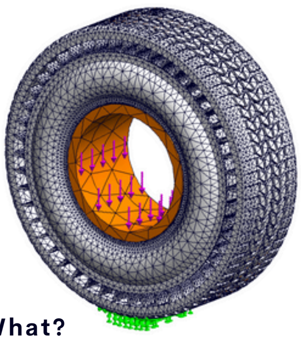
- The outcome of this project were **highly accurate 3D models** of **GTCP 36-150RJ** and **131-9 APUs**, achieved with a remarkable **90% fidelity** to the original.
- Developed **proficiency in CAD design** using **Solidworks 2023**, attention to detail, and ability to deliver precision in complex **Aerospace mechanical designs**



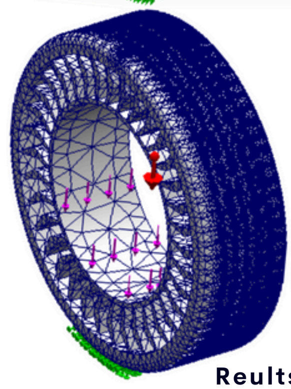


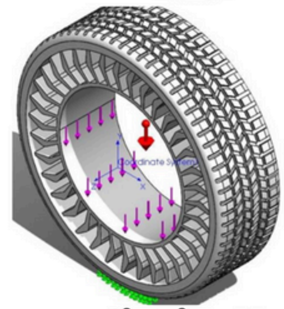


INNOVATION OF HYBRID TIRE TECHNOLOGY - NUST RESEARCH PAPER



Hybrid Tire Model, Simulation & Boundary Conditions





Non-Pneumatic Tire Model, Simulation & Boundary Conditions

What?

- **Invented** and Developed Hybrid Tire Technology
- Research Paper written on **Design, Analysis and Comparison of Hybrid and Non-Pneumatic Tires**
- **Hybrid tire** uses **NPTs puncture free** and **non-NPTs comfort** and **reduced weight** strengths
- Applicability for various applications and long-life

How?

- Used **Solidworks** for Designing of **Wheel** and **FEA Simulations** according to DFA/DFM
- **Theoretical research of Rim, spokes Structure Polyurethane Composite** material selection
- Applied **2 Different FEAs (static structural** and **Dynamic Analysis** on **Hybrid** and **NPT** tire Model as illustrated in **Figures**
- Applied **Bearing Distributed load of 5000N, Gravity** and Air Inflation **25psi Pressure** load
- Used **Jacobian 16 points** mesh nodes

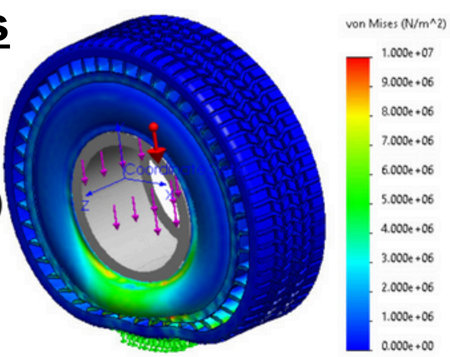
Reults?

- Successfully **proved Hybrid Tire's superiority over NPTs**
- Resulted in **more comfort** having more displacement over Impact loading
- Hybrid Tire has **30% weight reduction**
- **NPT Tires fail** under Impact loading conditions **shown by Equivalent Strain** were **Hybrid Tire don't fail** ensuring **better structural** integrity
- ***Research Document available link***

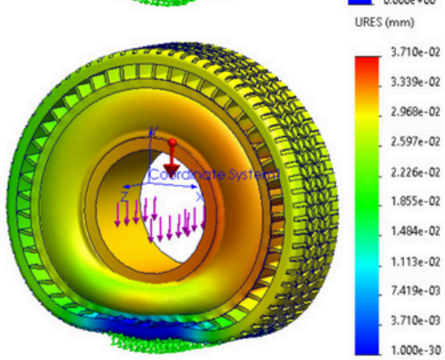
STATIC STRUCTURAL FINITE ELEMENT ANALYSIS FOR HYBRID TIRE SHOWN IN 1ST SIMULATION AND FOR NPT TIRE SHOWN IN 2ND SIMULATION FOR DESIGN VERIFICATION (DVP&R)

Hybrid Results

Von Mises Contours (MPa)

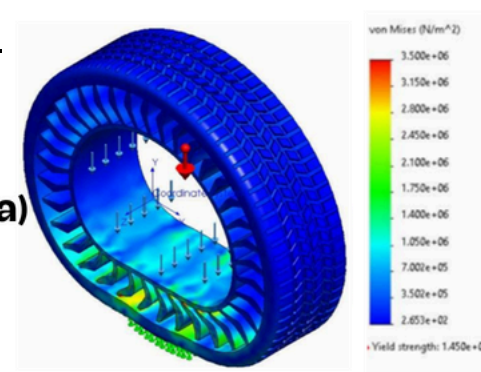


Displacement Contours(mm)

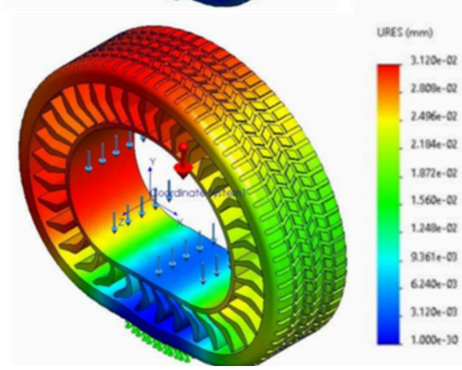


NPT Results

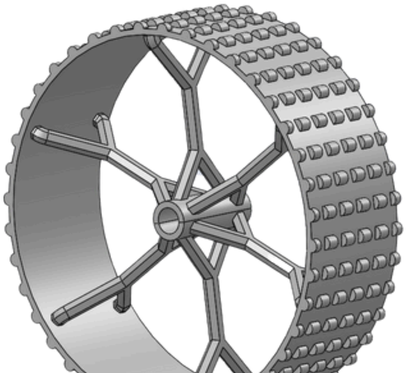
Von Mises Contours (MPa)

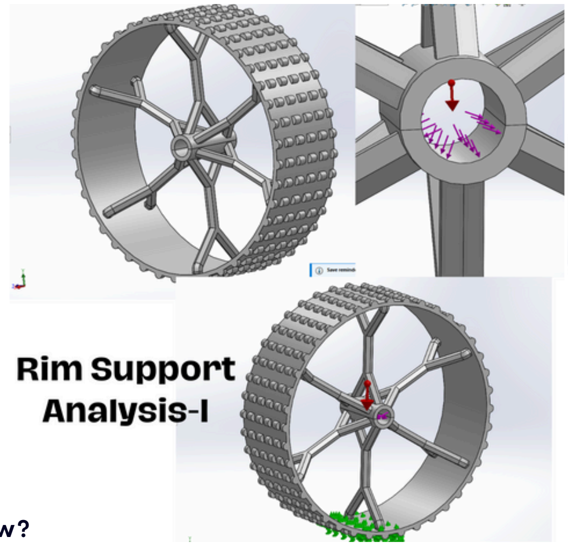


Displacement Contours(mm)

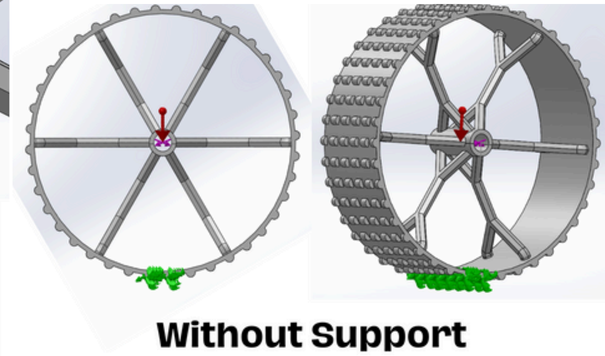


1ST IN SPACE ROVER WHEEL - PIAM 3D (NCOP, PAKISTAN)





Rim Support Analysis-I



Without Support Analysis-II

What?

- **PIAM 3D Pakistan's 2nd** Mechanical Design Competition
- **Designing** and **Rapid Prototyping** of **Space Rover Wheel** with **Min. Weight and Deflectio**n in range of **1 to 5mm**
- A **3D printed model** was **experimentally loaded** upon **8Kg load** for **Verification** and **Rankings**

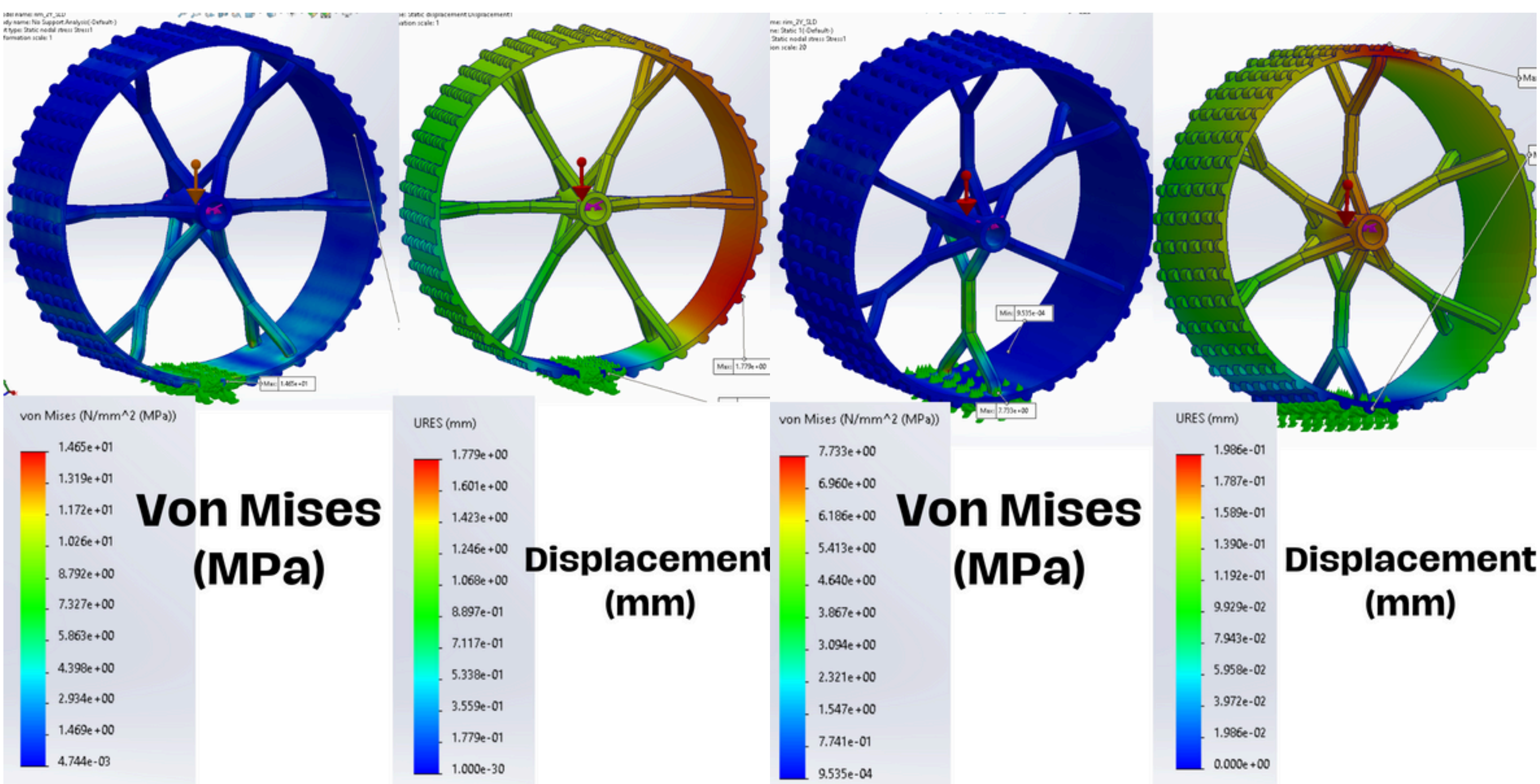
How?

- Used **Solidworks** for Designing of **Wheel** and **FEA Simulations** for **rigorous design optimizations** according to DFA/DFM
- Applied **GD&T** for **3D Printing** by **FDM 3D Printer** using **ABS plastic** material
- **Theoretical research of Rim Structure** and selected **Hexagonal C-S, Double Y shape rim**
- Applied **2 Different FEAs** on **Rim and Base** as well as **between Rim Spookes and Base** as illustrated in **Figures**
- Applied **Parabolic Distributed load of 8Kg**

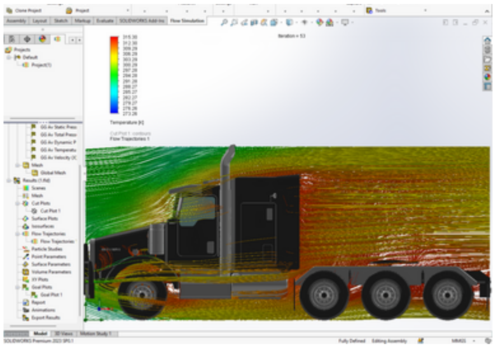
Reults?

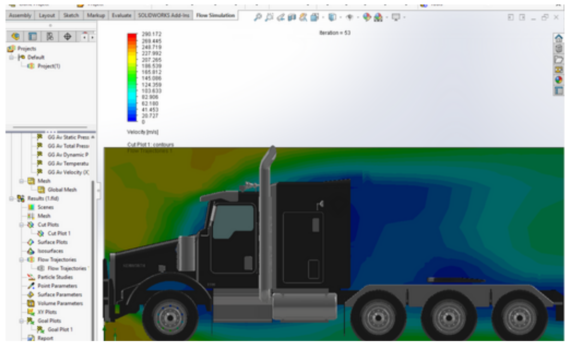
- Secured **1st position** in the Competition **Nationwide** Winning **1st prize money and certificate** amoung **38 teams**
- Achieved **Min. Weight of 75g** and **Min. Deflection of 1.8mm** upon **8Kg of applied loading as per competition rules**
- Innovative Design development and showed **Mechanics of Materials theoretical** and **computaional FEA** knowledge

FINITE ELEMENT ANALYSIS FOR WHEEL STRUCTURE WHILE LOADING APPLIED IN 1ST SIMULATION IS WITHOUT SPOKE SUPPORT AND IN 2ND SIMULATION WITH SPOKE FOR DESIGN OPTIMIZATION



CFD ANALYSIS OF KENWORTH T800 TRUCK - NUST FLUID MECHANICS





What?

- **Conducted CFD** analysis on **Kenworth T800 truck** for Fluid Mechanics 2 course, studying **airflow patterns** and **aerodynamics**.

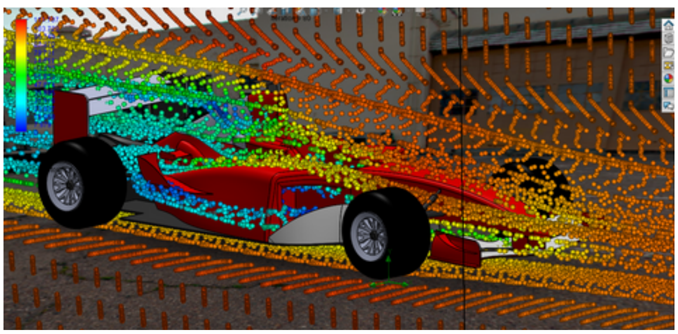
How?

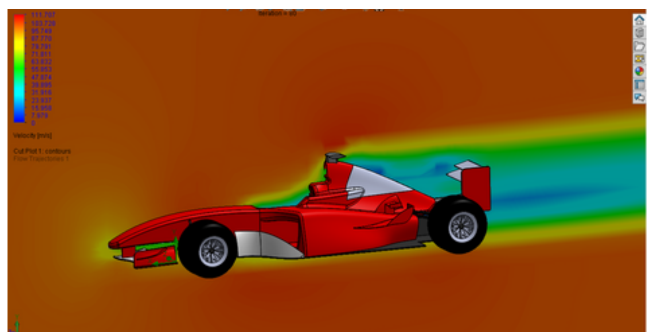
- Utilized **Solidworks Flow Simulation 2023** to **simulate airflow** interactions, setting up virtual environment with accurate **boundary conditions**.

Results?

- Achieved **15% reduced drag coefficient** through optimized **design modifications**.
- Calculated **pressure distributions**.
- Velocity profiles showed streamlined airflow, potentially **increasing fuel efficiency** by **4%**

CFD ANALYSIS OF FORMULA 1 VEHICLE - NUST FLUID MECHANICS COURSE





What?

- Calculated and the Demonstrated the **Pressure**, **Velocity** and **Temprature changes** of modified **profiles** on F1
- **Introduction of Aerodynamic Profiles** on vehicle and studying their effects

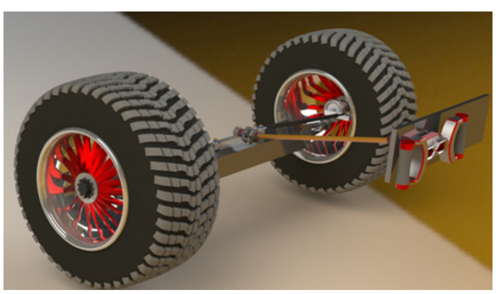
How?

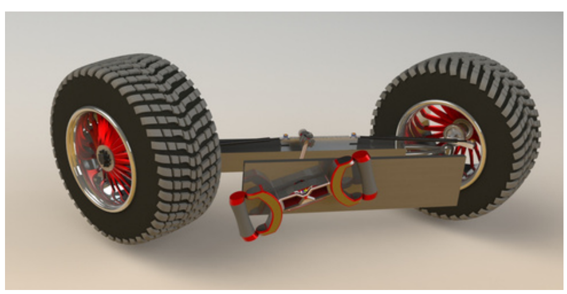
- Used **Solidworks** for Modifications using **Advanced CAD techniques**
- Performed **CFD Analysis** in **Solidworks Flow Simulation** using **Air** as Medium at **120kph** velocity

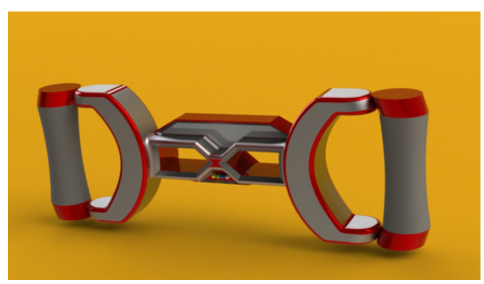
Results?

- **Reduced Wake regions**
- **Reduced** the **Drag** initiated by Wake using **Aerodynamic features** by **8%**
- **Studied** the **Aerodynamics** and **Fluid Dynamics**

RACK & PINION STEERING SYSTEM ASSEMBLY - NUST CAD COURSE







What?

- Desgining of a **Top Level Assembly** of Working of **Steering System** with **Rack & Pinion Mechanism**

How?

- Used **Solidworks** to design about **30+** individual **parts** and **5 Sub-Assemblies** while Considering **Real** World **Sizes** and **Dimensioning**.

Results?

- Demonstrated the complex mechanism
- Developed **New Design** of **Steering Wheel**
- Declared as **Best Semester Project**

SYED ALIYAR SHAH

OVERVIEW OF PROJECT AND RESEARCH PORTFOLIO DOCUMENT

WRITTEN PROJECTS AND RESEARCH/REPORTS

Ohmitron Battery Pack, Chargers and VPMS Provisional Patents and Proposals

- Patented Ohmitron's Li-ion Battery Packs, Chargers, VPMS and converters under US Patents
- Proposed the BPs as **Lead Engineer of Ohmitron and Battery Pack Design Team** to initiate manufacturing and commercializtion of BPs.

Design, Analysis and Comparison of NPTs and Hybrid Tires Research Paper:

- Innovation of Hybrid Tire made of Composite Materials and comparison with current Non-Pneumatic Tires

Review Paper on Extreme Conditions Composites and their Applications:

- Detalied review paper on performance of Composite Materials under Extreme Conditions in Nuclear, Automobile, Aerospace and Hydrogen Storage System and Industries

ICRAI Assessment of Impact Resistance Characteristics of 3D-Printed Polymers:

- Investigation of Impact assessments of 3D Printed Polymers including PLA-WM, PLA and ABS based on average damage area of fracture

Trailer Stability System Research Project:

- Trailer Stability System Project proposal to U.S DOT based on Vehicle Dynamics

Battery Materials Research:

- Produced an **Animated Video** on **basis of Project Report,** of Battery Materials and Induced ways of increasing Efficiency of **Li-Ion** and **Solid State Batteries**
- Used Video Editing Skills in **Filmora**, published a **Video** on **New Battery Materials** managing all group members

Shear Loading in Steel Bolted Beams

- Performed CAE using Finite Element Analysis to determine critical facet of bridge design
- Obtaining values under critical Design Factor of 2 for all 4 Bolts over a Lap Joint assembly and Theoretical Proof of calculations using Machine Design Techniques **(Project Link)**

Brinell's Hardness Test on Different Materials

- Performed Brinell's Hardness Test on **Cast Iron, Mild Steel** and **High Speed Steel (Project link)**
- Using **UHT Machine** and **2.5mm dia Carbide ball indentor** and **1839N Load for 10s**
- Obtained Brinell's Hardness Number (**BHN**) of **202, 150 & 287** for **Cast Iron, MS & HSS**

CAREER PROJECTS

- Completion of **numerous (50+) projects** in **Academic and Professional career** involving **curricular, extra-curricular, Governmental, National and International Compeititions Projects and awards**. Most projects aren't mentioned with some highlighted below:

Design and Development of Glass Fiber Reinforced Composite (GFRP) Insulators for HV Power Transmission:

- **Final Year Project** of development of **E-Glass Fiber Reinforced Polymer Matrix Composite (GFRPs with Epoxy Resign Matrix)** and **HTV Silicon Rubber** based Insulators for **36kV** rated Power Transmission Lines and Grid Stations
- Reduces Power Loss and Grounding, Electric Short Circuit Incidents compared to Ceramic Insulators with **200% longer life** using composite materials application
- **Fully Sponsored** Project by Composite Material **Industry (Polymer Tek PVT.LMT)**

Battery Pack Frame Fatigue Life Estimation:

- Estimation of Fatigue and Design Life of **Steel based Battery Pack Frame** for Trailer Chasis Mounted Electric Trucks
- Used **Isotropic Materials** Static, Dynamic and Frequency **FE Analyses**

Battery Pack Cell Holder Fatigue/Design life Simulation Analysis:

- Estimation of Battery Pack Cell Holder Fatigue and Design Life made of **3D Printed Polymer (ABS)** and **Composite CFRP Material** for structural integrity.
- Utilized Static, Frequency and Dynamic Loads FE Simulations

Ohmitron's propietrary MegaWatt Charging System:
Ohmitron's Propietrary Vehicle Power Management System (VPMS)
Ohmitron's Propietrary SiC based Single Phase Power Converters:

Human Powered Vehicle

- Designed a Working Vehicle solely powered by Human Power, **Consuming Zero Fuel** and adding to sustainable development and green technology by using **Solidworks** and Engineering Drawing, the vehicle is in **fabrication** stage.

Arduino-Based Battery Management & Monitoring System

- Developed a Battery Management System prototype for **real-time individual cell parametres like Voltages, Current, SOC and Cell Balancing (Project Link)**
- Used **Arduino for Cell Monitoring & Management, ESP32 for wireless communication and Monitoring through Blynk Server and BM3451 for Cell Balancing.** Successfully achieved project objectives, providing **real-time LCD Display and Online Cell Monitoring accurately** contributing to intelligent energy storage and sustainable solutions.

CFD Analysis of F1 Vehicle, Kenworth T800 Truck & Heat Exchanger

- Performed **Computational Fluid Dynamics** on **F1, Kenworth T800 Truck and Counter Flow Heat Exchangers** using **Solidworks**
- Calculated **Theoretical Values** for Pressure, Velocity, Temprature and **Induced Extruded Profiles** on the Bodies for **Reducing Drag, improving Aerodynamics & Heat Transfer**

Mercedez F1 Vehicle and Steering System Design

- Recreated a **precise** and **intricate** model of the **Mercedes Formula 1** vehicle for **Dassault Systems** and a Complete Steering System in **Solidworks** and **PTC CREO** as a **Top level Assembly** with **100% real-time systematics.**
- Executed the project in **15 focused on-screen hours over 5 days**
- **Dassault Systems Showcased** The Vehicle **on Media Platforms**